JOAQUIN MAIDANA BORJAS

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EDUCATION

Loras College, Dubuque, IA
Bachelor of Science in Engineering and Computer Science | Expected May 2029
GPA: 3.88

SKILLS

Languages: Spanish: Fluent | English: Fluent | French: Intermediate-high | Mandarin: Intermediate-low
Technical: SolidWorks Simulations: Intermediate | OpenFOAM | ParaView | Python | C++ | HTML-CSS | Arduino IDE | SQL | Linux

PRESENTATIONS

AI-Guided Navigation and Decision-Making for Robots
Iowa Space Grant Consortium (ISGC/ NASA-Funded) Spring Research Symposium, Drake University, April 2026
Loras Legacy Symposium Poster Presentation, Loras College, April 2026

PROJECTS

Loras College Engineering Club, Dubuque, IA
Mobile Aerial Reconnaissance System (MARS) funded by ISGC/NASA | September 2026 – Present

- Co-designed and helped lead development of MARS, a tethered autonomous drone-rover system integrating computer vision, terrain mapping, and Theta* path planning for cooperative aerial-ground navigation
- Architected a three-subsystem robotics pipeline (aerial image collection, rover-based computation, tether-assisted communication) to offload real-time perception and control to a central ground unit
- Implemented autonomous navigation algorithms including weighted occupancy grid mapping, slope-constrained cost evaluation, and spline-based path smoothing
- Technologies: Python, C++, Machine Learning, SolidWorks, Embedded Systems, Ubuntu

Battle Bots Autonomous Robot funded by ISGC/NASA | January 2026 – Present

- Design and implement neural network models to optimize robot decision-making in competitive battle scenarios
- Develop mechanical components in SolidWorks and run simulation-based analyses to guide design improvements
- Integrate hardware sensors with software control systems for real-time autonomous navigation and combat strategies
- Technologies: Python, Neural Networks, SolidWorks, Embedded Systems

EXPERIENCE

Loras College, Dubuque, IA
AV Student Worker | March 2026 – Present
Duhawk Dunkers Coach | October 2025 – Present

ACTIVITIES

Loras Men's Basketball | 2025 – Present
Honors Program | 2025 – Present
ISA Prep Academy Basketball Team Captain | 2024 – 2025